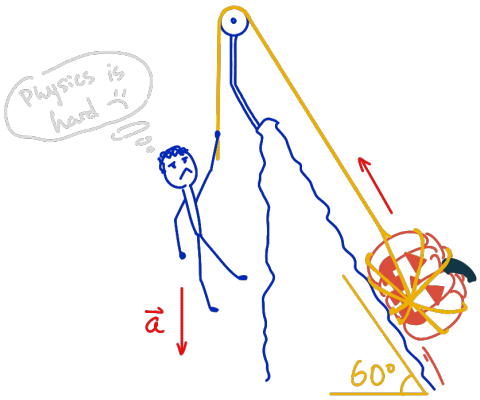
Physics 5A Midterm 2 Review Session

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Use $g = 9.8\,\text{m/s}^2$ where needed. For notation, I will usually denote $\vec{g} = (0,\, -9.8)\,\text{m/s}^2$ for gravitational acceleration, in the conventional coordinate system.

Problem 1: Jinyan hangs from a rope with his feet off the ground. The rope is connected to a gigantic pumpkin on an incline. The incline makes an angle $60°$ with the horizontal. The rope passes over a fixed, massless, frictionless pulley. The rope is massless, stays taut, and does not stretch. Jinyan's mass is $m_J = 60\,\text{kg}$, and the pumpkin's mass is $m_P = 10\,\text{kg}$. At the instant we consider, Jinyan is moving downward and the pumpkin is sliding uphill without rolling. The coefficient of kinetic friction between the pumpkin and the incline is $\mu_k = 0.10$. Neglect air resistance.



Part (a): Draw the free-body diagrams for Jinyan and the pumpkin. Indicate the directions of all the forces.

Part (b): For Jinyan, take downward to be positive. For the pumpkin, take uphill to be the positive parallel direction and away from the surface to be the positive perpendicular direction. Find the parallel and perpendicular components of the gravitational force on the pumpkin. Then compute the normal force and kinetic friction acting on it.

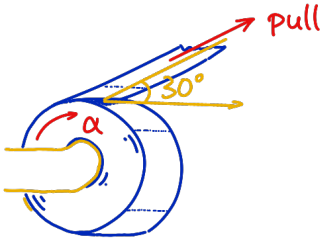
Part (c): Explain why the two objects have the same acceleration magnitude and why the tension has the same magnitude on both sides of the pulley. Write Newton's second law separately for Jinyan and for the pumpkin along their directions of motion.

Part (d): Solve your two equations obtained in part (c) to find the acceleration of each object and the tension T.

Problem 2: Jinyan is pulling toilet paper from a cylindrical toilet paper roll of mass M and outer radius R. The roll rotates about a fixed, frictionless horizontal axle through its center. Viewed along the axle, the paper leaves the roll tangentially, and Jinyan pulls it to the right at an angle 30° above the horizontal with a constant force of magnitude F. The paper is taut and does not slip on the roll.

The roll starts from rest at initial time. During a measured time interval of t seconds, it rotates through an angular displacement $\theta > 0$, measured in radians. Take clockwise rotation to be positive. During this short interval, treat M, R, and the moment of inertia I as constant, and neglect the mass of the paper pulled off the roll. Also neglect friction and any resisting torque from separating the paper layers.

We write the unknown moment of inertia about the axle as $I = cMR^2$. We want to determine c from the measured motion, without assuming the roll is a uniform solid cylinder. Treat F, M, R, t, and θ as known quantities specified above, so that your final answer should be written in terms of (some or all of) these quantities.



Part (a): Draw the lever arm vector $\vec{r}$ from the axle to the point where the paper leaves the roll, and the pulling force $\vec{F}$ at that point. Find the angle ϕ between these two vectors. Then find the net torque about the axle. Explain why gravity and the axle's supporting force do not contribute torque about this axis.

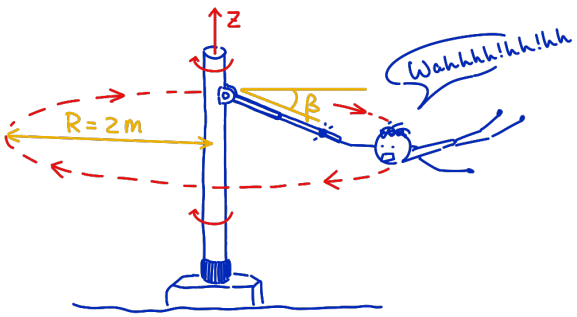
Part (b): Use the measured angular displacement θ and elapsed time t to find the angular acceleration α of the toilet paper roll.

Part (c): Use the net torque and angular acceleration obtained in parts (a) and (b) to find I, then find the coefficient c in terms of F, M, R, t, and θ.

Part (d): Find the final angular velocity ω_f at the end of the time interval.

Problem 3: A vertical shaft rotates about the z-axis. A long rigid rod of adjustable length is attached to the shaft by a hinge located on the z-axis. The hinge turns with the shaft but allows the rod to tilt freely up or down. Jinyan holds the other end of the rod with his feet off the ground. As the shaft turns, Jinyan moves around the z-axis in a horizontal circle.

Treat Jinyan as a point mass $m = 60\,\text{kg}$ at the outer end of the rod. Neglect the mass of the rod and air friction. Consider uniform circular motion: the rod has adjustable length such that Jinyan undergoes circular motion around the z-axis with a fixed radius $R = 2\,\text{m}$. For different rotation speeds, one will observe different tilts (up or down) of the rod. Let β be the angle that the rod makes below the horizontal, measured from the hinge toward Jinyan, with $0 < \beta < 90°$. Initially, Jinyan moves at a tangential speed $v = 4.00\,\text{m/s}$.



Part (a): Draw the free-body diagram for Jinyan (when he gets to the right-most point as shown in the figure). Denote the force that Jinyan exerts on the rod while holding it as $-\vec{F}_\text{h}$. Using the free-body diagram, and the fact that Jinyan is undergoing circular motion, explain why the rod's tilt angle depends on Jinyan's tangential speed in his circular motion.

Part (b): Write Newton's second law for Jinyan's body, in the vertical direction and in the horizontal direction toward the center of the circle. Then compute the vector-component form of the force $\vec{F}_\text{h}$ that the rod exerts on Jinyan.

Part (c): Compute the angle β, measured clockwise below the horizontal as drawn.

Part (d): If the vertical shaft rotates too fast, Jinyan will not be able to hold on to the rod. The maximal amount of force that Jinyan can hold the rod is 1000 N. Find his maximal tangential speed for him to maintain the circular motion, and the corresponding angle β. Assume that, when β is changing, the radius of the circular path stays fixed at $R = 2\,\text{m}$.